Dhanashree Somani
ASU ID: 1239782599
CSE 575: Statistical Machine Learning | Naive Bayes Classifier Project

Project Report — Naive Bayes Classifier

1. Introduction

This project implements a Naive Bayes classifier on MNIST dataset to differentiate between digits '0' and '1'. It follows the supervised learning approach where two hand-engineered features (fit using Maximum Likelihood Estimation (MLE) to Gaussian distributions) of each image are used to classify unseen test images using Bayes theorem with uniformly distributed class priors.

2. Task 1: Feature Extraction

Each of the 28x28 grayscale image is flattened to a 784-dimensional pixel vector. Computing two scalar features for each image:

- **Feature 1 — Mean brightness:** the average pixel intensity across all of the 784 pixels. Digit '0' images consist of a hollow ring structure that produces mean brightness higher than digit '1'.
- **Feature 2 — Std deviation of brightness:** pixel spread is measured. Digit '0' consist of both dark background and bright foreground pixels are arranged in a ring, with a higher standard deviation than digit '1'.

These two features are selected because they capture mutually exclusive structural differences between the two digit classes and is easier to compute without any preprocessing.

3. Task 2: Parameter Estimation — MLE

Under the assumption of the Naive Bayes, the features are assumed to be separate 1-D Gaussian. MLE in the case of a Gaussian gives out a sample mean and a sample variance. The eight personalized training set parameters are estimated.

(Student ID: 2599) are summarised below:

Parameter	Digit 0	Digit 1
Mean — Feature 1 (average brightness)	44.17	19.34
Variance — Feature 1	116.44	31.18
Mean — Feature 2 (std brightness)	87.38	61.30
Variance — Feature 2	102.67	82.57

The huge variation in mean brightness (i.e. 44.17 and 19.34), hence supporting good classifier performance as the two classes are well-separated in feature space.

4. Task 3: Naive Bayes Classification

The log-posterior is computed for each class for each test image. As we have assumed equal priors (P(Y=0) = P(Y=1) = 0.5), the log-prior term is cancelled and the decision reduces to comparing log-likelihoods. The log-likelihoods are determined as the sum of two log-Gaussian functions one of which is part of each feature and is used to indicate the Naive Bayes assumption of conditional independence. The Naive Bayes independence assumption is reasonable here, because mean and standard deviation capture different aspects of image structure.

5. Task 4: Results and Accuracy

Test Set	Total Images	Correct	Accuracy
Digit 0	980	899	91.73%
Digit 1	1135	1048	92.33%

6. Observations & Analysis

The classifier achieves over 91% accuracy for both test sets using only two simple pixel-level.

Critical observations:

- **Discriminability of features:** The prime determinant of the classifying ability is the large difference between the mean brightness of the digit 0 (44.17) and the brightness of the digit 1 (19.34). The space occupied by digit 0 in the image (the circular ring) is larger, and digit 1 is a thin vertical line, and it indicates that there are more dark pixels.
- **Variance analysis:** Both features also exhibit greater variance in Digit 0 indicates that it has a more complicated spatial arrangement. Hence, it is a bit more difficult to model accurately pointed by the minorly reduced accuracy when compared with digit 1 (91.73 ~ 92.33).
- **Misclassifications**: The error rate is about 8% and is mostly due to unusually written samples, example a very thin 0 can appear with low mean brightness in the area of a 1, or a bold 1 can appear with high mean brightness in the area of 0.
- **Independence assumption:** Naive Bayes independence assumption of features is an approximation - the mean and standard deviation are related to each other up to some extent. This relationship can be well-represented in a full 2-D Gaussian model having a covariance term and hence can be more accurate.
- **Prior assumption:** Equal priors were assumed, in spite the fact that the test sets were not of the same size (980 vs 1135). The decision boundary can change slightly while using observed priors that are based on the recurrence of the training set.

7. Conclusion

This project depicts that a Naive Bayes classifier with MLE-estimated Gaussian parameters can be highly effective in separating two digits, 0 and 1, in MNIST by using only two image-level features. The accuracy of ~92% confirms the approach and highlights how intuitive feature engineering combined with a principled probabilistic framework can yield competitive performance on a binary classification task.